

Marketing Campaign Performance Analysis

Facebook Ads vs. AdWords

A/B Testing, Statistical Analysis & Conversion Insights

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SQL • Python • Power BI • Excel • Data Visualization • Statistical Analysis

Project Overview

This project analyzes the performance of two digital advertising platforms, **Facebook Ads** and **AdWords**, using daily marketing campaign data.

The analysis focuses on campaign performance across **clicks, conversions, conversion rates, advertising costs, and cost efficiency** to identify differences between the two platforms.

The project combines **Exploratory Data Analysis (EDA), correlation analysis, hypothesis testing, linear regression, and time-based analysis** to generate data-driven insights that can support marketing budget allocation and campaign optimization.

Business Problem

As a marketing agency, our primary objective is to maximize the return on investment (ROI) for our clients' advertising campaigns.

In this project, we analyze two advertising platforms — **Facebook Ads and AdWords** — to determine which platform delivers better results in terms of **clicks, conversions, conversion rate, and overall cost-effectiveness**.

By comparing campaign performance and applying statistical analysis, the objective is to identify meaningful differences between the two platforms and determine where marketing resources can potentially be allocated more effectively.

The findings can help support data-driven decisions for improving campaign performance and optimizing future advertising strategies.

Business Objectives

The primary objectives of this project are to:

1. Compare the overall performance of Facebook Ads and AdWords.
2. Analyze clicks and conversions generated by each platform.
3. Compare conversion rates between the two campaigns.
4. Examine advertising costs and cost per conversion.
5. Analyze the relationship between clicks and conversions.
6. Determine whether the difference in conversion performance is statistically significant.
7. Examine the relationship between advertising spend and campaign outcomes.
8. Analyze campaign performance across different months and days of the week.
9. Identify actionable insights that can support marketing budget allocation and campaign optimization.

Dataset Overview

The dataset contains daily marketing campaign performance data for **Facebook Ads and AdWords**.

Dataset Information

Attribute	Details
Records	365 daily observations
Features	17 columns
Campaigns	Facebook Ads & AdWords
Analysis Period	January 2019 – December 2019
Key Metrics	Views, Clicks, Conversions, CTR, Conversion Rate, Cost
Analysis Type	Comparative Campaign Analysis & Statistical Analysis

Key Performance Metrics

- **Ad Views**
- **Ad Clicks**
- **Ad Conversions**
- **Click-Through Rate (CTR)**
- **Conversion Rate**
- **Advertising Cost**
- **Cost per Conversion**

Tools & Technologies

Category	Tools
Programming Language	Python
Data Manipulation	Pandas, NumPy

Data Visualization	Matplotlib, Seaborn
Statistical Analysis	SciPy
Machine Learning	Scikit-learn
Development Environment	Jupyter Notebook

Analytical Techniques

- Exploratory Data Analysis (EDA)
- Descriptive Statistics
- Correlation Analysis
- Hypothesis Testing
- Independent Samples T-Test
- Linear Regression
- Time-Series Analysis
- Campaign Performance Comparison

Research Question

Which ad platform is more effective in terms of conversions, clicks, and overall cost-effectiveness?

Importing Libraries

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import scipy.stats as st
import numpy as np
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score, mean_squared_error
from statsmodels.tsa.seasonal import seasonal_decompose
from statsmodels.tsa.stattools import coint
import warnings
warnings.filterwarnings('ignore')
```

Data Description

The dataset contains daily performance data for two different advertising campaigns conducted during the year **2019**. The campaigns represent **Facebook Ads** and **AdWords Ads**, allowing their performance to be compared across the year.

Each row represents the campaign performance recorded for a particular day. Since the dataset covers the complete year from **January 1st, 2019, to December 31st, 2019**, it

contains a total of **365 daily records**. The dataset includes several performance indicators that help evaluate the effectiveness and efficiency of both advertising campaigns over time.

Key Features Included in the Dataset

- **Date:** The date associated with each campaign record, covering the period from January 1st, 2019, to December 31st, 2019.
- **Ad Views:** The total number of times an advertisement was viewed.
- **Ad Clicks:** The number of clicks generated by the advertisement.
- **Ad Conversions:** The number of conversions generated as a result of the advertisement.
- **Cost per Ad:** The amount spent on running the respective advertising campaign.
- **Click-Through Rate (CTR):** The proportion of ad views that resulted in clicks, indicating how effectively the advertisement attracts user engagement.
- **Conversion Rate:** The proportion of clicks that resulted in conversions, showing how effectively the campaign turns clicks into desired actions.
- **Cost per Click (CPC):** The average amount spent for each click generated by the advertisement.

Loading the dataset

```
df = pd.read_csv('marketing_campaign.csv')
```

data overview

```
df.head()
```

	Date	Facebook Ad Campaign	Facebook Ad Views	Facebook Ad Clicks	\
0	1/1/2019	FB_Jan19	2116	18	
1	1/2/2019	FB_Jan19	3106	36	
2	1/3/2019	FB_Jan19	3105	26	
3	1/4/2019	FB_Jan19	1107	27	
4	1/5/2019	FB_Jan19	1317	15	

	Facebook Ad Conversions	Cost per Facebook Ad	\
0	8	\$126	
1	12	\$104	
2	8	\$102	
3	9	\$71	
4	7	\$78	

	Facebook Click-Through Rate (Clicks / View)	\
0	0.83%	
1	1.15%	
2	0.84%	
3	2.45%	
4	1.10%	

	Facebook Conversion Rate (Conversions / Clicks)	\
0	42.73%	
1	34.04%	

2	31.45%
3	34.76%
4	47.59%

	Facebook Cost per Click (Ad Cost / Clicks)	AdWords Ad Campaign \
0	\$7.14	AW_Jan19
1	\$2.91	AW_Jan19
2	\$3.89	AW_Jan19
3	\$2.62	AW_Jan19
4	\$5.38	AW_Jan19

	AdWords Ad Views	AdWords Ad Clicks	AdWords Ad Conversions \
0	4984	59	5
1	4022	71	6
2	3863	44	4
3	3911	49	5
4	4070	55	7

	Cost per AdWords Ad	AdWords Click-Through Rate (Clicks / View) \
0	\$194	1.18%
1	\$75	1.77%
2	\$141	1.13%
3	\$141	1.26%
4	\$133	1.36%

	AdWords Conversion Rate (Conversions / Click) \
0	8.40%
1	7.80%
2	9.59%
3	11.08%
4	12.22%

	AdWords Cost per Click (Ad Cost / Clicks)
0	\$3.30
1	\$1.05
2	\$3.23
3	\$2.86
4	\$2.40

rows and columns count of the dataset

df.shape

(365, 17)

datatypes of the columns

df.dtypes

Date	object
Facebook Ad Campaign	object
Facebook Ad Views	int64

```

Facebook Ad Clicks                int64
Facebook Ad Conversions            int64
Cost per Facebook Ad               object
Facebook Click-Through Rate (Clicks / View) object
Facebook Conversion Rate (Conversions / Clicks) object
Facebook Cost per Click (Ad Cost / Clicks) object
AdWords Ad Campaign                object
AdWords Ad Views                   int64
AdWords Ad Clicks                  int64
AdWords Ad Conversions              int64
Cost per AdWords Ad                object
AdWords Click-Through Rate (Clicks / View) object
AdWords Conversion Rate (Conversions / Click) object
AdWords Cost per Click (Ad Cost / Clicks) object
dtype: object

```

```
# converting date to datetime
```

```
df['Date'] = pd.to_datetime(df['Date'])
```

```
# descriptive stats of the campaigns
```

```
df.describe()
```

	Date	Facebook Ad Views	Facebook Ad Clicks \
count	365	365.000000	365.000000
mean	2019-07-02 00:00:00	2179.687671	44.049315
min	2019-01-01 00:00:00	1050.000000	15.000000
25%	2019-04-02 00:00:00	1656.000000	35.000000
50%	2019-07-02 00:00:00	2202.000000	43.000000
75%	2019-10-01 00:00:00	2717.000000	54.000000
max	2019-12-31 00:00:00	3320.000000	73.000000
std	NaN	618.074639	12.140559

	Facebook Ad Conversions	AdWords Ad Views	AdWords Ad Clicks \
count	365.000000	365.000000	365.000000
mean	11.742466	4717.19726	60.383562
min	5.000000	3714.00000	31.000000
25%	10.000000	4247.00000	49.000000
50%	12.000000	4711.00000	60.000000
75%	13.000000	5190.00000	73.000000
max	19.000000	5760.00000	89.000000
std	2.924786	561.11406	14.368225

	AdWords Ad Conversions
count	365.000000
mean	5.980822
min	3.000000
25%	5.000000
50%	6.000000
75%	7.000000

```
max          9.000000
std          1.628106
```

Comparing Campaigns performance

Distribution of clicks and conversions

Facebook Ads

```
plt.figure(figsize=(15,6))

plt.subplot(1,2,1)
plt.title('Facebook Ad Clicks', color='#1F4E78', fontsize=14,
fontweight='bold')
sns.histplot(
    df['Facebook Ad Clicks'],
    bins=7,
    color='#2F75B5',
    edgecolor='#1F4E78',
    kde=True,
    line_kws={'color': '#008C95', 'linewidth': 2}
)

plt.subplot(1,2,2)
plt.title('Facebook Ad Conversions', color='#1F4E78', fontsize=14,
fontweight='bold')
sns.histplot(
    df['Facebook Ad Conversions'],
    bins=7,
    color='#2F75B5',
    edgecolor='#1F4E78',
    kde=True,
    line_kws={'color': '#008C95', 'linewidth': 2}
)

plt.tight_layout()
plt.show()
```

AdWords

```
plt.figure(figsize=(15,6))

plt.subplot(1,2,1)
plt.title('AdWords Ad Clicks', color='#1F4E78', fontsize=14,
fontweight='bold')
sns.histplot(
    df['AdWords Ad Clicks'],
    bins=7,
    color='#008C95',
    edgecolor='#1F4E78',
```

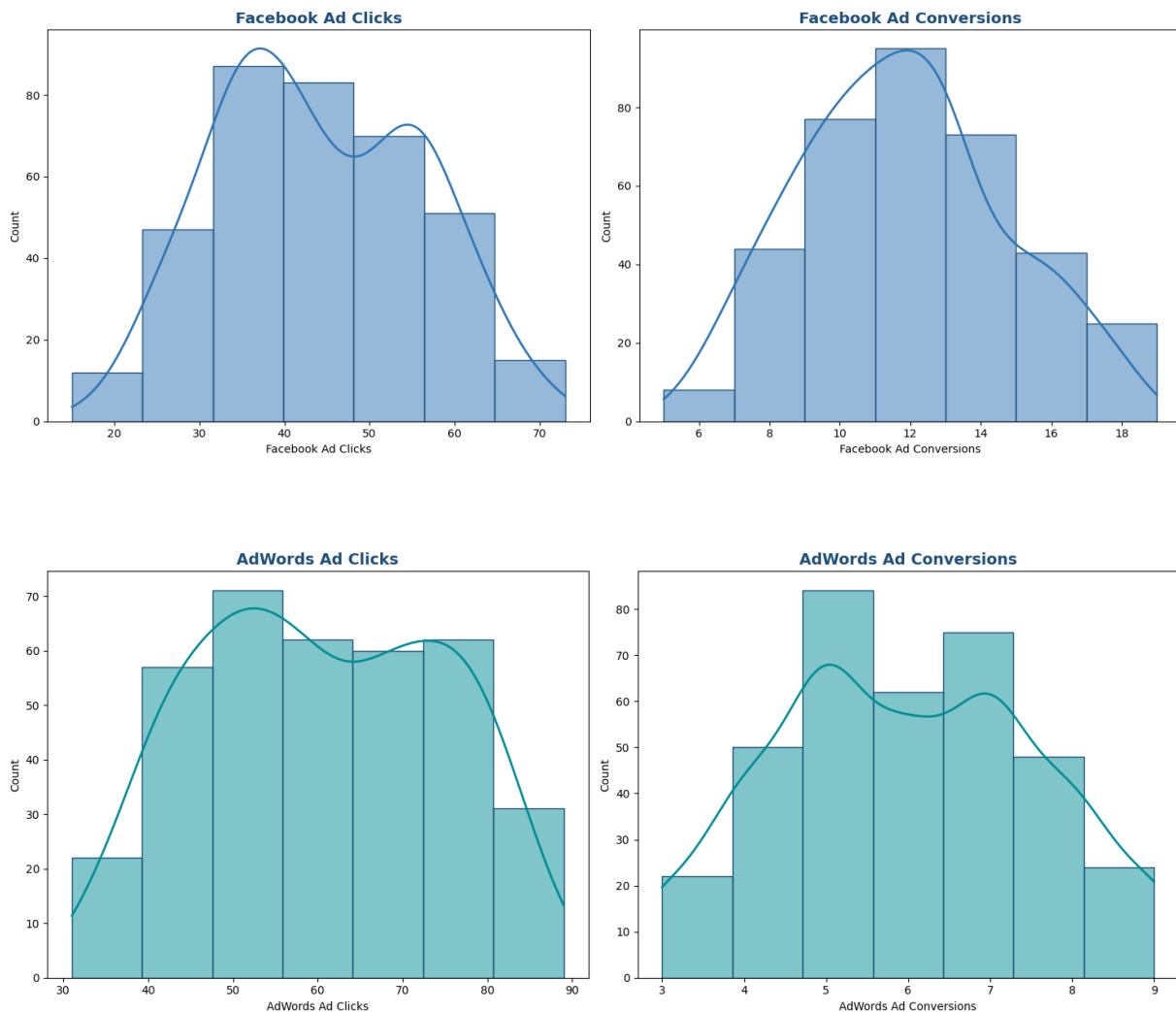
```

    kde=True,
    line_kws={'color': '#2F75B5', 'linewidth': 2}
)

plt.subplot(1,2,2)
plt.title('AdWords Ad Conversions', color='#1F4E78', fontsize=14,
fontweight='bold')
sns.histplot(
    df['AdWords Ad Conversions'],
    bins=7,
    color='#008C95',
    edgecolor='#1F4E78',
    kde=True,
    line_kws={'color': '#2F75B5', 'linewidth': 2}
)

plt.tight_layout()
plt.show()

```



The histograms exhibit a relatively symmetrical distribution, indicating that clicks and conversions are fairly evenly distributed across the observed values. This suggests that there are no substantial extreme values or outliers at either the higher or lower end of the distributions.

How frequently do we observe days with high numbers of conversions compared to days with low numbers of conversions?

creating function to calculate the category for the conversions

```
def create_conversion_category(conversion_col):
    category = []
    for conversion in df[conversion_col]:
        if conversion < 6:
            category.append('less than 6')
        elif 6 <= conversion < 11:
            category.append('6 - 10')
        elif 11 <= conversion < 16:
            category.append('10 - 15')
        else:
            category.append('more than 15')
    return category
```

applying function of different campaign's conversions

```
df['Facebook Conversion Category'] = create_conversion_category('Facebook Ad
Conversions')
df['AdWords Conversion Category'] = create_conversion_category('AdWords Ad
Conversions')
```

```
df[['Facebook Ad Conversions', 'Facebook Conversion Category', 'AdWords Ad
Conversions', 'AdWords Conversion Category']].head()
```

	Facebook Ad Conversions	Facebook Conversion Category	\
0	8	6 - 10	
1	12	10 - 15	
2	8	6 - 10	
3	9	6 - 10	
4	7	6 - 10	

	AdWords Ad Conversions	AdWords Conversion Category
0	5	less than 6
1	6	6 - 10
2	4	less than 6
3	5	less than 6
4	7	6 - 10

```
df['Facebook Conversion Category'].value_counts()
```

Facebook Conversion Category	
10 - 15	189
6 - 10	128

```
more than 15      47
less than 6        1
Name: count, dtype: int64
```

```
facebook = pd.DataFrame(df['Facebook Conversion
Category'].value_counts()).reset_index().rename(columns = {'Facebook
Conversion Category': 'Category'})
facebook
```

	Category	count
0	10 - 15	189
1	6 - 10	128
2	more than 15	47
3	less than 6	1

```
df['AdWords Conversion Category'].value_counts()
```

```
AdWords Conversion Category
6 - 10      209
less than 6  156
Name: count, dtype: int64
```

```
adwords = pd.DataFrame(df['AdWords Conversion
Category'].value_counts()).reset_index().rename(columns = {'AdWords
Conversion Category': 'Category'})
adwords
```

	Category	count
0	6 - 10	209
1	less than 6	156

```
category_df = pd.merge(facebook, adwords, on = 'Category', how =
'outer').fillna(0)
category_df
```

	Category	count_x	count_y
0	10 - 15	189	0.0
1	6 - 10	128	209.0
2	less than 6	1	156.0
3	more than 15	47	0.0

```
category_df = category_df.iloc[[3,1,0,2]]
category_df
```

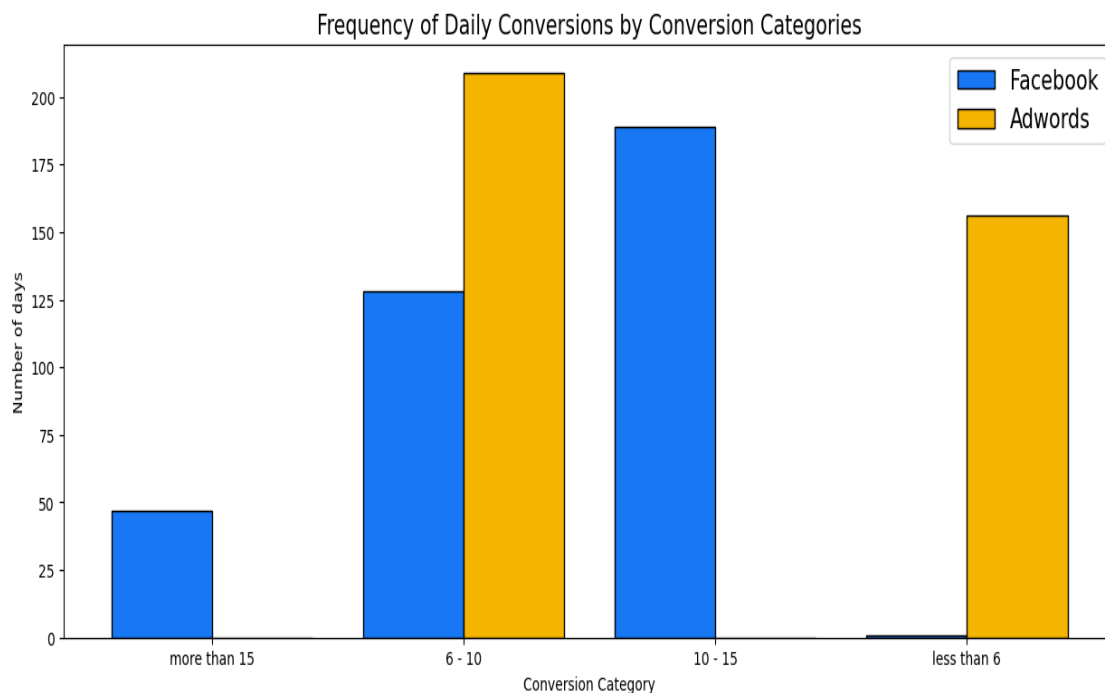
	Category	count_x	count_y
3	more than 15	47	0.0
1	6 - 10	128	209.0
0	10 - 15	189	0.0
2	less than 6	1	156.0

```
X_axis = np.arange(len(category_df))
X_axis
```

```
array([0, 1, 2, 3])
```

```
X_axis = np.arange(len(category_df))
plt.figure(figsize = (15,6))
plt.bar(X_axis - 0.2, category_df['count_x'], 0.4, label = 'Facebook', color = '#1877F2', linewidth = 1, edgecolor = 'k')
plt.bar(X_axis + 0.2, category_df['count_y'], 0.4, label = 'Adwords', color = '#F4B400', linewidth = 1, edgecolor = 'k')

plt.xticks(X_axis, category_df['Category'])
plt.xlabel("Conversion Category")
plt.ylabel("Number of days")
plt.title("Frequency of Daily Conversions by Conversion Categories", fontsize = 15)
plt.legend(fontsize = 15)
plt.show()
```

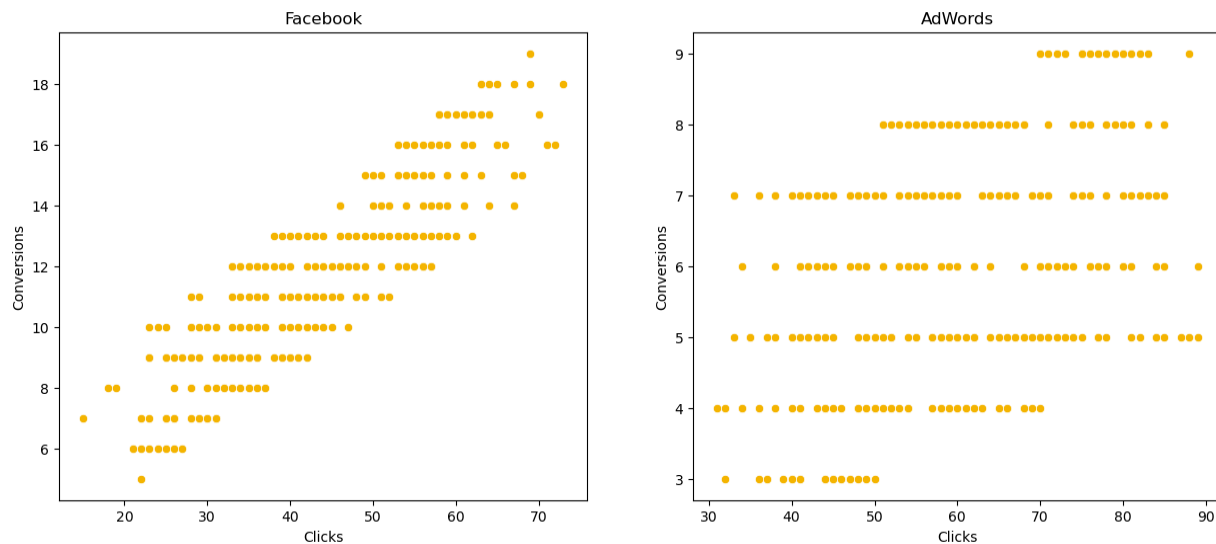


- The data suggests Facebook had more frequent higher conversion days than AdWords, which either had very low conversion rates (less than 6) or moderate ones (6 - 10).
- There is a significant variance in the number of high-conversion days between two different campaigns.
- The absence of any days with conversions between 10 - 15 and more than 15 in AdWords indicates a need to review what strategies were changed or what external factors could have influenced these numbers.

Do more clicks on the ad really lead to more sales?

```
plt.figure(figsize=(15,6))
plt.subplot(1,2,1)
```

```
plt.title('Facebook')
sns.scatterplot(x = df['Facebook Ad Clicks'],y = df['Facebook Ad
Conversions'], color = '#F4B400')
plt.xlabel('Clicks')
plt.ylabel('Conversions')
plt.subplot(1,2,2)
plt.title('AdWords')
sns.scatterplot(x = df['AdWords Ad Clicks'],y = df['AdWords Ad Conversions'],
color = '#F4B400')
plt.xlabel('Clicks')
plt.ylabel('Conversions')
plt.show()
```



```
facebook_corr = df[['Facebook Ad Conversions','Facebook Ad Clicks']].corr()
facebook_corr
```

	Facebook Ad Conversions	Facebook Ad Clicks
Facebook Ad Conversions	1.000000	0.873775
Facebook Ad Clicks	0.873775	1.000000

```
adwords_corr = df[['AdWords Ad Conversions','AdWords Ad Clicks']].corr()
adwords_corr
```

	AdWords Ad Conversions	AdWords Ad Clicks
AdWords Ad Conversions	1.000000	0.447993
AdWords Ad Clicks	0.447993	1.000000

```
print('Correlation Coeff \n-----')
print('Facebook : ',round(facebook_corr.values[0,1],2))
print('AdWords : ',round(adwords_corr.values[0,1],2))
```

```
Correlation Coeff
-----
```

Facebook : 0.87
AdWords : 0.45

- A correlation coefficient of **0.87** indicates a strong positive linear relationship between Facebook ad clicks and sales. This means that an increase in Facebook ad clicks is generally associated with an increase in sales.
- The strong correlation indicates that Facebook ad clicks have a significant relationship with sales and may explain a substantial portion of the variation observed in sales performance.
- The strong positive relationship between Facebook ad clicks and sales suggests that Facebook advertising can play an important role in generating sales. Increasing investment in Facebook advertising or improving campaign performance could potentially contribute to higher sales.
- A correlation coefficient of **0.45** indicates a moderate positive linear relationship between AdWords ad clicks and sales. This shows that AdWords clicks are positively associated with sales, although the relationship is weaker compared with Facebook ads.
- The moderate correlation suggests that AdWords advertising contributes to sales, but its impact may also depend on other factors. Further analysis could help identify these factors and support better optimization of AdWords campaigns.

Hypothesis Testing

Hypothesis: Advertising on Facebook will result in a greater number of conversions compared to advertising on AdWords.

Null Hypothesis (H0): There is no difference in the number of conversions between Facebook and AdWords, or the number of conversions from AdWords is greater than or equal to those from Facebook.

H0: $\mu_{\text{Facebook}} \leq \mu_{\text{AdWords}}$

Alternate Hypothesis (H1): The number of conversions from Facebook is greater than the number of conversions from AdWords.

H1: $\mu_{\text{Facebook}} > \mu_{\text{AdWords}}$

```
print('Mean Conversion \n-----')
print('Facebook :', round(df['Facebook Ad Conversions'].mean(),2))
print('AdWords :', round(df['AdWords Ad Conversions'].mean(),2))
```

```
t_stats, p_value = st.ttest_ind(a = df['Facebook Ad Conversions'], b =
df['AdWords Ad Conversions'], equal_var = False)
print('\nT statistic', t_stats, '\np-value',p_value)
```

```
# comparing the p value with the significance of 5% or 0.05
if p_value < 0.05:
    print("\np-value is less than significance value, Reject the null hypothesis")
else:
    print("\np-value is greater than significance value, Accept the null hypothesis")
```

Mean Conversion

Facebook : 11.74

AdWords : 5.98

T statistic 32.88402060758184

p-value 9.348918164530465e-134

p-value is less than significance value, Reject the null hypothesis

- The mean number of conversions from Facebook ads (11.74) is substantially higher than the mean number of conversions from AdWords ads (5.98). This suggests that, on average, Facebook advertising is more effective in generating conversions compared to AdWords advertising.
- The T statistic (32.88) is a measure of the difference between the means of the two groups relative to the variation within the groups. A larger T statistic indicates a greater difference between the means of the two groups.
- The p-value (9.35e-134) is extremely small, indicating strong evidence against the null hypothesis.
- The results strongly support the alternate hypothesis, indicating that the number of conversions from Facebook advertising is indeed greater than the number of conversions from AdWords advertising.
- Facebook advertising appears to be a more effective channel for generating conversions compared to AdWords advertising, based on the sample data analyzed.
- Given the significant difference in conversion rates between Facebook and AdWords, consider reallocating resources towards Facebook advertising efforts. This could involve increasing ad spend, expanding targeting efforts, or experimenting with different ad formats to capitalize on the platform's effectiveness in driving conversions.

Regression Analysis

What will happen when I do go with the Facebook Ad? How many facebook ad conversions can I expect given a certain number of facebook ad clicks?

independent variable

```
X = df[['Facebook Ad Clicks']]
```

dependent variable

```
y = df[['Facebook Ad Conversions']]
```

```
# initializing and fitting Linear Regression model
```

```
reg_model = LinearRegression()  
reg_model.fit(X,y)  
prediction = reg_model.predict(X)
```

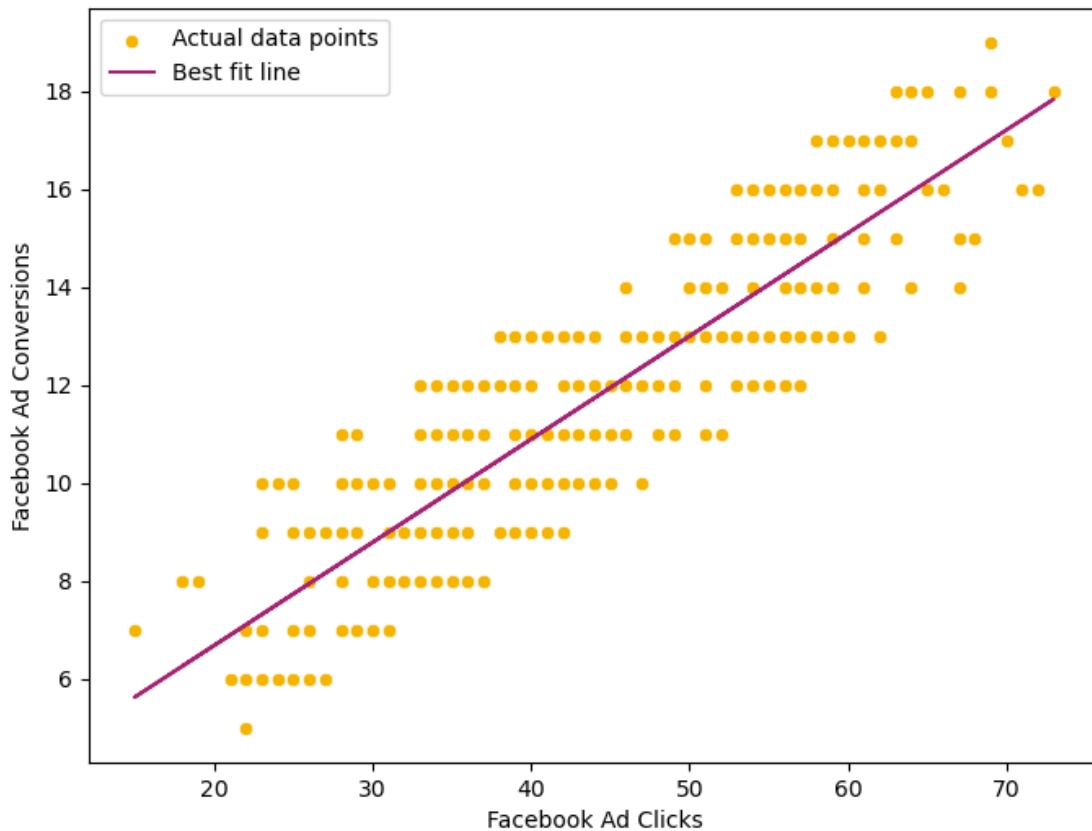
```
# model evaluation
```

```
r2 = r2_score(y, prediction)*100  
mse = mean_squared_error(y, prediction)  
print('Accuracy (R2 Score):',round(r2,2),'%')  
print('Mean Squared Error:', round(mse,2))
```

Accuracy (R2 Score): 76.35 %

Mean Squared Error: 2.02

```
plt.figure(figsize=(8,6))  
sns.scatterplot(x = df['Facebook Ad Clicks'],y = df['Facebook Ad  
Conversions'], color = '#F4B400', label = 'Actual data points')  
plt.plot(df['Facebook Ad Clicks'], prediction, color = '#A62372', label =  
'Best fit line')  
plt.legend()  
plt.show()
```



```
print(f'For {50} Clicks, Expected Conversion :  
{round(reg_model.predict([[50]])[0][0],2)}')
```

```
print(f'For {80} Clicks, Expected Conversion :  
{round(reg_model.predict([[80]])[0][0],2)}')
```

For 50 Clicks, Expected Conversion : 13.0

For 80 Clicks, Expected Conversion : 19.31

- The model has a reasonably good predictive power, with an R2 score of 76.35%. This suggests that it can effectively predict Facebook ad conversions based on the number of Facebook ad clicks.
- With the insights provided by the Linear Regression model, businesses can make informed decisions about resource allocation, budget planning, and campaign optimization.
- For instance, knowing the expected number of Facebook ad conversions based on a certain number of Facebook ad clicks can help in setting realistic campaign goals, optimizing ad spend, and assessing the ROI of Facebook advertising efforts.

Analyzing Facebook Campaign metrics over time.

cleaning data (removing unwanted symbols from the columns and converting them to numerical columns)

```
df['Facebook Click-Through Rate (Clicks / View)'] = df['Facebook Click-Through Rate (Clicks / View)'].apply(lambda x: float(x[:-1]))
df['Facebook Conversion Rate (Conversions / Clicks)'] = df['Facebook Conversion Rate (Conversions / Clicks)'].apply(lambda x: float(x[:-1]))
df['Facebook Cost per Click (Ad Cost / Clicks)'] = df['Facebook Cost per Click (Ad Cost / Clicks)'].apply(lambda x: float(x[1:]))
df['Cost per Facebook Ad'] = df['Cost per Facebook Ad'].apply(lambda x: float(x[1:]))
```

filtering for facebook campaign

```
df = df[['Date', 'Facebook Ad Views',  
        'Facebook Ad Clicks', 'Facebook Ad Conversions', 'Cost per Facebook Ad',  
        'Facebook Click-Through Rate (Clicks / View)',  
        'Facebook Conversion Rate (Conversions / Clicks)',  
        'Facebook Cost per Click (Ad Cost / Clicks)']]
```

```
df.head()
```

	Date	Facebook Ad Views	Facebook Ad Clicks	Facebook Ad Conversions
0	2019-01-01	2116	18	8
1	2019-01-02	3106	36	12
2	2019-01-03	3105	26	8
3	2019-01-04	1107	27	9
4	2019-01-05	1317	15	7

	Cost per Facebook Ad	Facebook Click-Through Rate (Clicks / View)
0	126.0	0.83

1	104.0	1.15
2	102.0	0.84
3	71.0	2.45
4	78.0	1.10

	Facebook Conversion Rate (Conversions / Clicks) \
0	42.73
1	34.04
2	31.45
3	34.76
4	47.59

	Facebook Cost per Click (Ad Cost / Clicks)
0	7.14
1	2.91
2	3.89
3	2.62
4	5.38

At what times of the month or days of the week do we observe the conversions?

extracting month and week day from the date column

```
df['month'] = df['Date'].dt.month
```

```
df['week'] = df['Date'].dt.weekday
```

```
plt.figure(figsize=(8,5))
```

```
plt.title('Weekly Conversions')
```

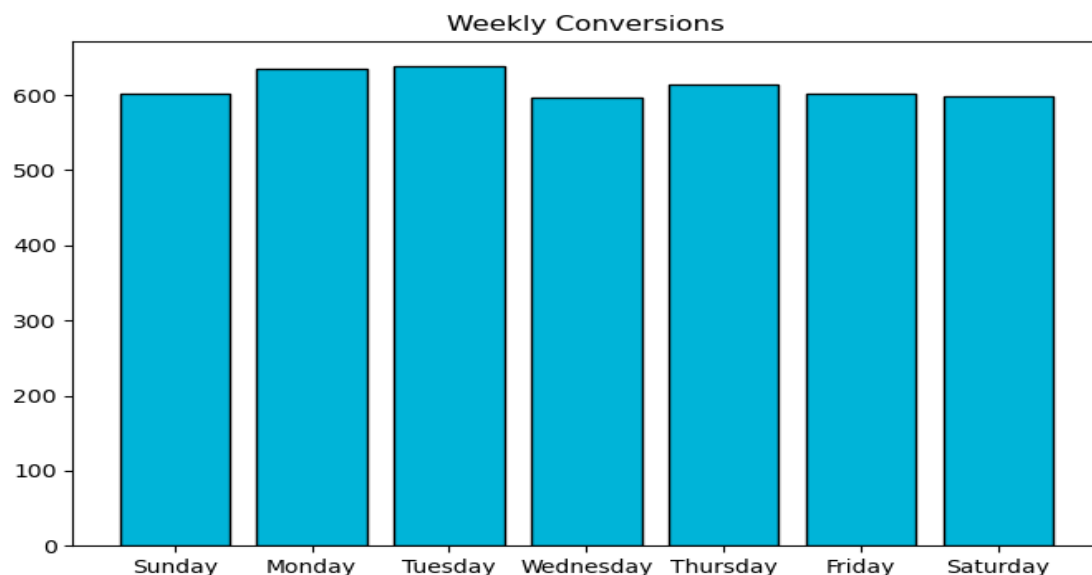
```
weekly_conversion = df.groupby('week')[['Facebook Ad Conversions']].sum()
```

```
week_names=
```

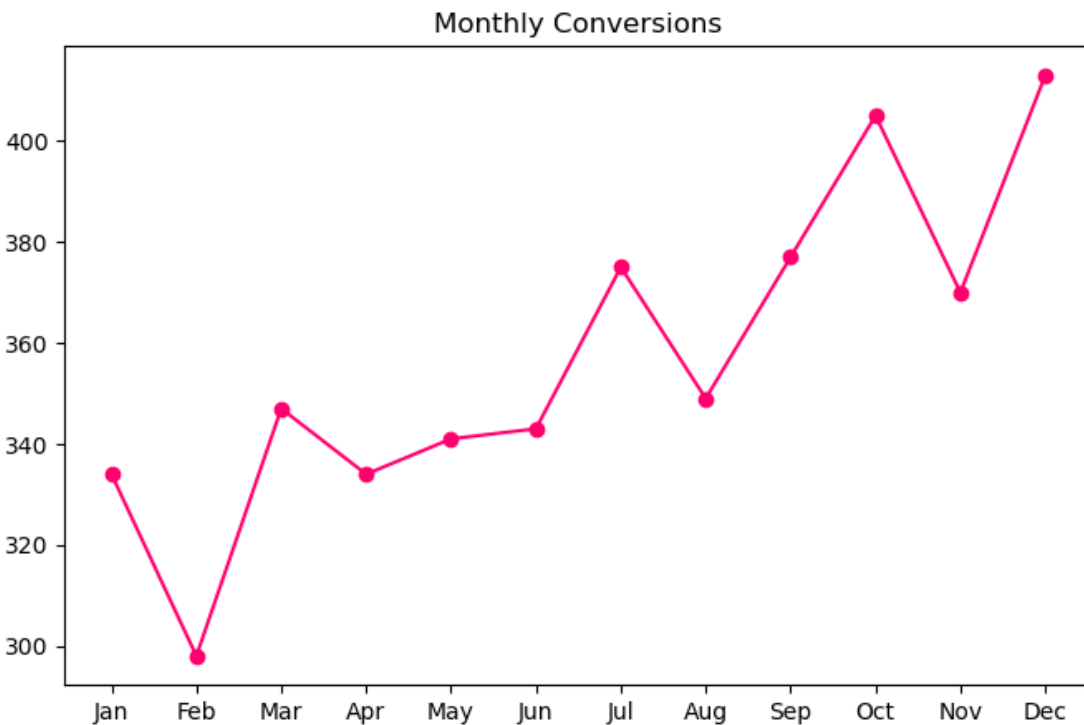
```
['Sunday', 'Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday', 'Saturday']
```

```
plt.bar(week_names, weekly_conversion['Facebook Ad Conversions'], color = '#00B4D8', edgecolor = 'k')
```

```
plt.show()
```



```
plt.figure(figsize=(8,5))
plt.title('Monthly Conversions')
monthly_conversion = df.groupby('month')[['Facebook Ad Conversions']].sum()
month_names =
['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun', 'Jul', 'Aug', 'Sep', 'Oct', 'Nov', 'Dec']
plt.plot(month_names, monthly_conversion['Facebook Ad Conversions'], '-o',
color = '#FF006E')
plt.show()
```



- The total number of conversions across weekdays remains fairly consistent throughout the year, suggesting a stable level of user engagement across the week. However, **Mondays and Tuesdays consistently record the highest conversion rates**, indicating that users may be more engaged or responsive to marketing activities at the beginning of the workweek.
- The monthly conversion trend shows an **overall upward pattern**, indicating a general growth in conversions over time. However, some months show noticeable fluctuations. **February, April, May, June, August, and November** experience lower conversions compared with the months immediately before or after them. These declines may be associated with factors such as **seasonal variations, changes in customer behavior, or modifications in marketing strategies**.

How does the Cost Per Conversion (CPC) trend over time?

Cost Per Conversion (CPC): This metric is used to evaluate the cost effectiveness and profitability of an online advertising campaign. This metric helps marketers understand

how much they are spending to obtain each conversion, allowing them to optimize their spending and targeting strategies effectively.

```
plt.figure(figsize=(8,5))
plt.title('Monthly Cost Per Conversion (CPC)')

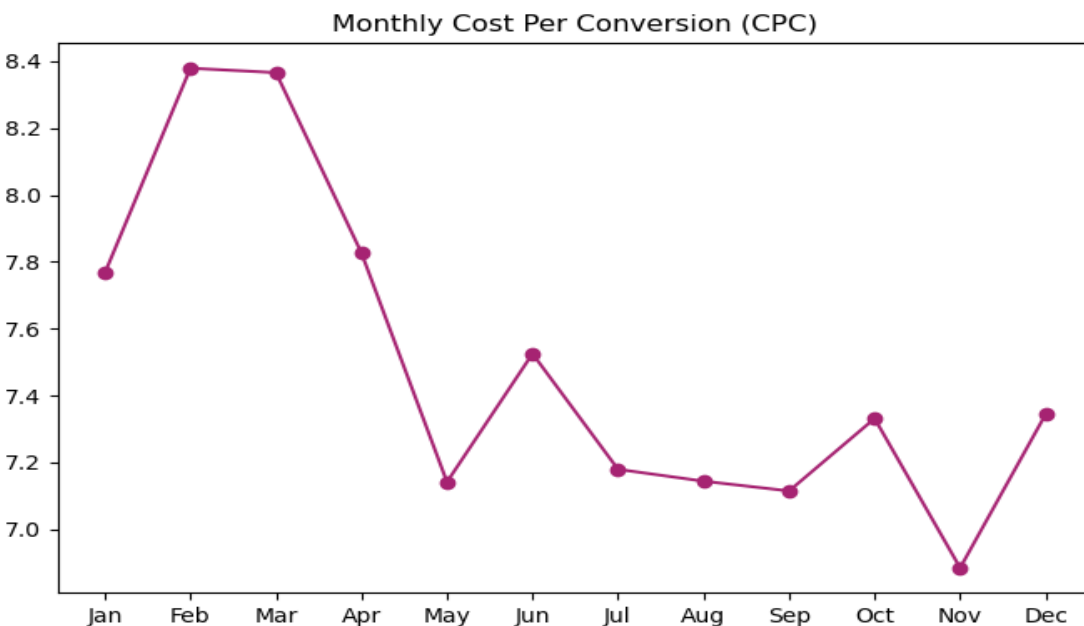
df['Cost per Facebook Ad'] = (
    df['Cost per Facebook Ad']
    .astype(str)
    .str.replace('$', '', regex=False)
    .str.replace(',', '', regex=False)
    .astype(float)
)

monthly_df = df.groupby('month')[['Facebook Ad Conversions', 'Cost per Facebook Ad']].sum()

monthly_df['Cost per Conversion'] = (
    monthly_df['Cost per Facebook Ad'] / monthly_df['Facebook Ad Conversions']
)

plt.plot(
    month_names,
    monthly_df['Cost per Conversion'],
    '-o',
    color='#A62372'
)

plt.show()
```



- The CPC trend across the 12-month period shows some variation, but overall, it remains within a relatively stable range.
- **May and November** record the lowest CPC values, suggesting that advertising may be more cost-effective or that conversion rates are higher during these months.
- **February** records the highest CPC value, indicating that advertising costs may be comparatively higher during this period.
- The lower CPC observed in months such as **May and November** may indicate stronger advertising efficiency or more favorable market conditions during these periods.
- To maximize ROI, the business could consider allocating a larger portion of the advertising budget to months with historically lower CPC values, particularly **May and November**.

Is there a long-term equilibrium relationship between advertising spend and conversion rates that suggests a stable, proportional impact of budget changes on conversions over time?

```
score, p_value, _ = coint(df['Cost per Facebook Ad'], df['Facebook Ad
Conversions'])
print('Cointegration test score:', score)
print('P-value:', p_value)
if p_value < 0.05:
    print("\np-value is less than significance value, Reject the null
hypothesis")
else:
    print("\np-value is greater than significance value, Accept the null
hypothesis")
```

Cointegration test score: -14.755428385103226

P-value: 2.1337375979060563e-26

p-value is less than significance value, Reject the null hypothesis

- Since the **p-value is significantly below the selected significance level**, we reject the null hypothesis. This provides evidence of a **long-term equilibrium relationship between advertising spend (cost) and conversions**.
- Understanding this stable relationship between cost and conversions can help businesses improve their advertising strategies. By focusing investment on campaigns that generate a strong **return on investment (ROI)** and adjusting budgets according to performance, companies can aim to **increase conversions while keeping costs under control**.

Key Findings

The analysis of Facebook Ads and AdWords produced several important findings.

Facebook Ads

- Facebook generated a higher average number of conversions than AdWords.
- Facebook clicks showed a strong positive relationship with conversions.
- The statistical analysis indicates a significant difference in conversion performance between the two platforms.
- Click volume can provide useful information for understanding Facebook conversion performance.

AdWords

- AdWords generated a lower average conversion level compared with Facebook in the analyzed dataset.
- The relationship between clicks and conversions was positive.
- The campaign provides opportunities for further optimization of targeting, messaging, and conversion strategy.

Statistical Findings

The hypothesis testing results indicate that the observed difference in conversion performance between Facebook Ads and AdWords is statistically significant based on the selected significance level.

Overall Finding: Facebook Ads demonstrated stronger conversion performance than AdWords in the analyzed dataset.

Business Recommendations

Based on the analysis, the following recommendations can be considered:

1. Optimize Budget Allocation

Consider prioritizing advertising platforms that demonstrate stronger conversion performance and better cost efficiency.

2. Strengthen Facebook Campaigns

Analyze the characteristics of high-performing Facebook campaigns and use these insights to improve audience targeting, ad creatives, and campaign strategy.

3. Improve AdWords Performance

Review AdWords targeting, keywords, messaging, and conversion strategy to identify opportunities for improving campaign effectiveness.

4. Monitor Conversion Efficiency

Track conversion rate and cost per conversion alongside clicks and impressions to evaluate campaign quality more effectively.

5. Use Time-Based Insights

Consider campaign timing and historical performance patterns when planning future advertising campaigns and allocating budgets.

6. Continue A/B Testing

Future experiments can test different audiences, creatives, messaging, offers, and landing pages to identify strategies that generate better conversion outcomes.

Conclusion

This project evaluated the performance of **Facebook Ads and AdWords** using exploratory analysis, statistical testing, correlation analysis, regression, and time-based analysis.

The analysis indicates that **Facebook Ads achieved stronger conversion performance than AdWords** within the analyzed dataset. The statistical testing further provides evidence that the observed difference in conversion performance is statistically significant.

The findings demonstrate how marketing campaign data can be transformed into actionable insights to support **data-driven campaign optimization, performance monitoring, and more informed advertising budget allocation**.

Overall, this analysis highlights the importance of combining business metrics with statistical evidence rather than relying solely on descriptive comparisons when evaluating marketing campaigns.
